

模式识别与机器学习

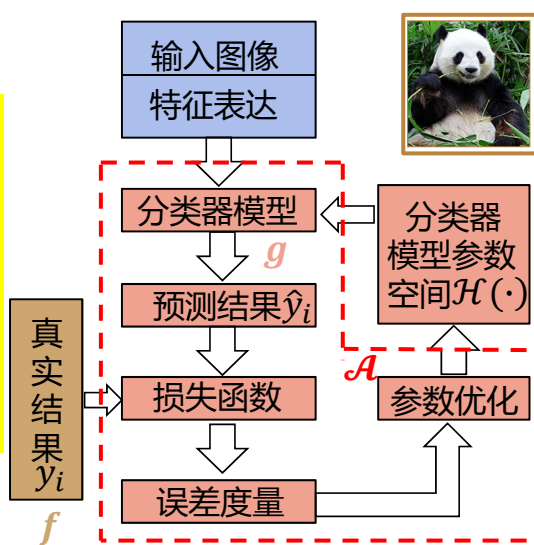
授课教师：杨卫东 胡静



1.2 模式识别方法与机器学习



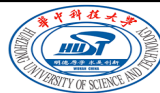
分类器的设计



算法 $\mathcal{A}$

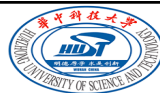
- ① 感知器
- ② 线性回归
- ③ Fisher线性判别
- ④ 逻辑斯蒂回归
- ⑤ 非线性变换
- ⑥ 线性支撑向量机
- ⑦ 对偶SVM与核SVM
- ⑧ 多类分类
- ⑨ 神经网络与深度学习
- ⑩ 卷积神经网络

第四讲 Fisher线性判别 (*Fisher Discriminant Analysis*)



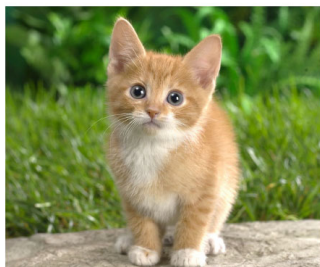
- 4.1 Fisher线性判别动机 (*The goal of Fisher Linear Discriminant*)
- 4.2 Fisher线性判别分析 (*Fisher Discriminant Analysis*)
- 4.3 Fisher线性判别算法 (*Fisher Discriminant Algorithm*)

3.1 线性回归问题



为什么需要回归?

分类(*Classification*)



输出结果(*Output*):
类别标签(Class
label)

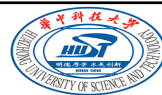
检测(分类+定位)

Detection (Classification + Localization)

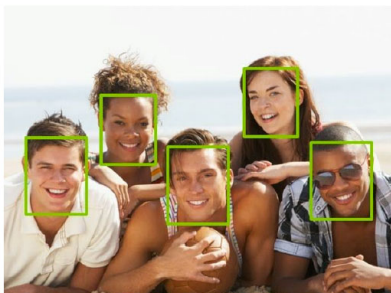


输出结果(*Output*): 类别标签+位置坐标
Class label + Box in the image (x, y,
w, h)

4.1 Fisher线性判别动机

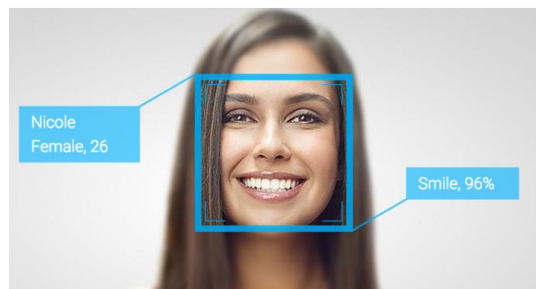


应用示例:



人脸检测

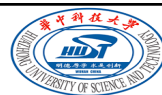
(Detection finds the faces in images)



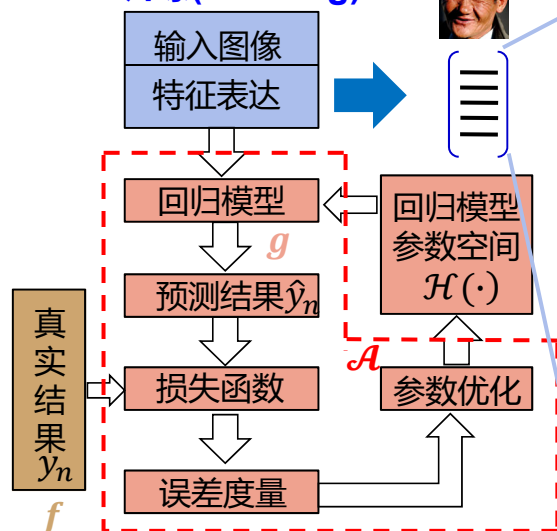
人脸识别

(Recognition recognizes WHO the person is)

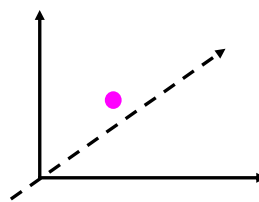
4.1 Fisher线性判别动机



训练(Training)

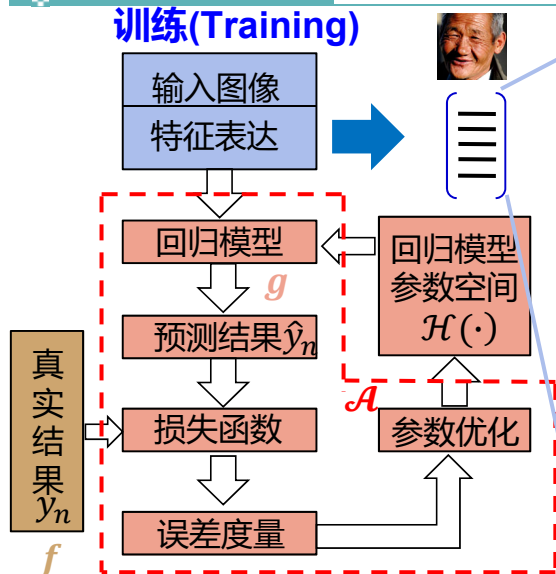


➤ 图像是特征空间中的一个“点”

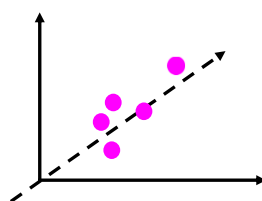
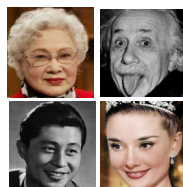


4.1 Fisher线性判别动机

训练(Training)

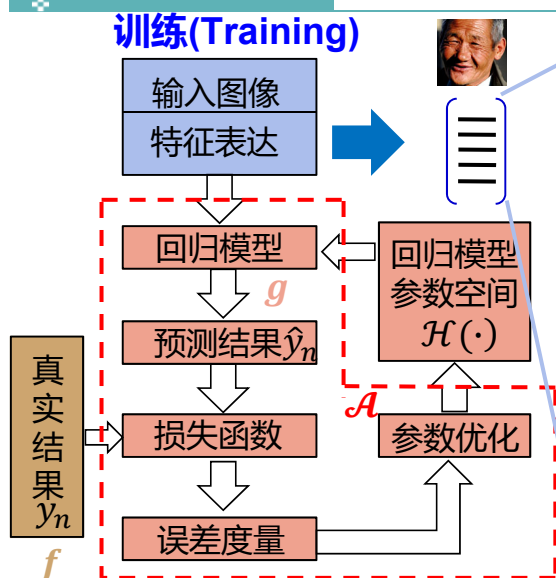


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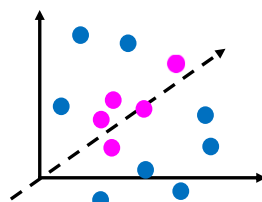
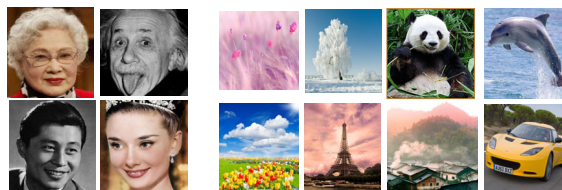


4.1 Fisher线性判别动机

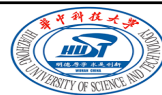
训练(Training)



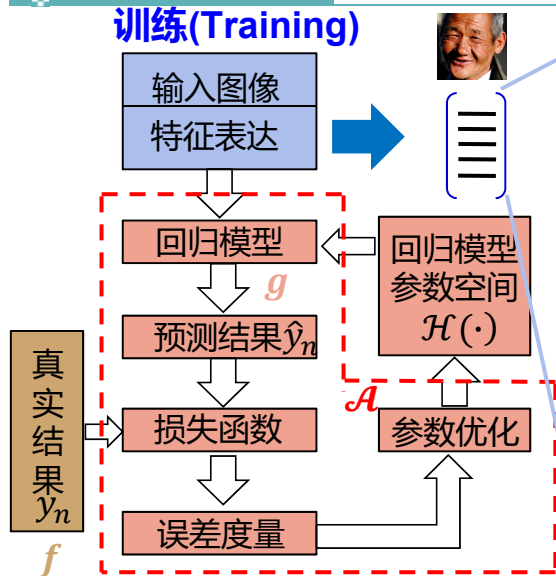
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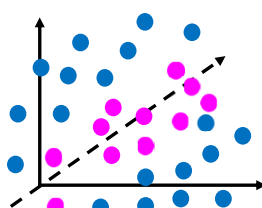
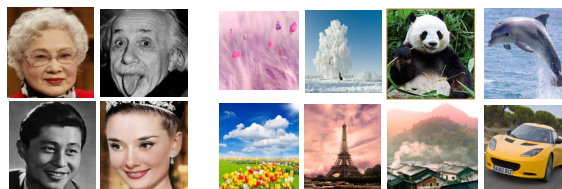
4.1 Fisher线性判别动机



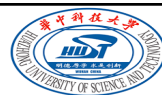
训练(Training)



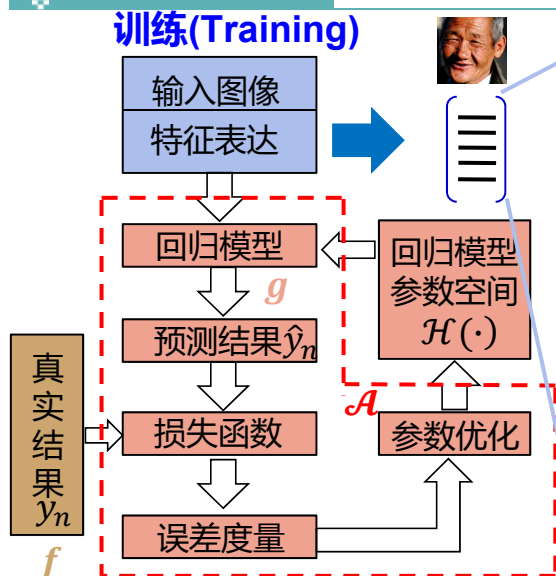
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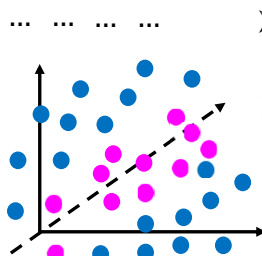
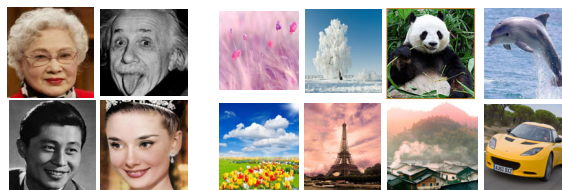
4.1 Fisher线性判别动机



训练(Training)



➤ 图像是特征空间中的一个“点”



➤ 高维特征空间：

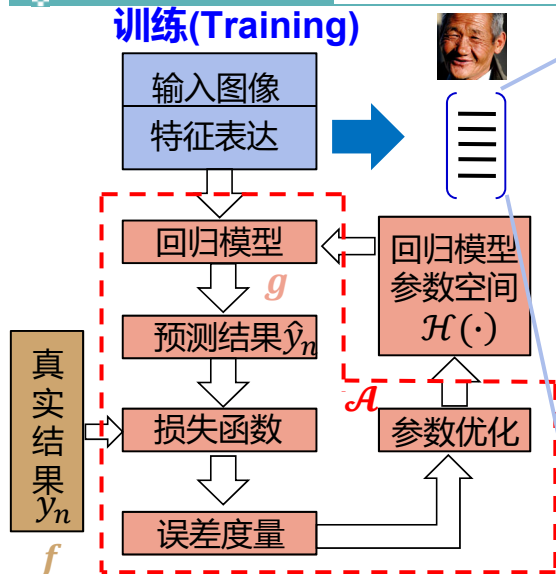
例如：100*100*3

➤ 人脸只是众多样本中的一部分

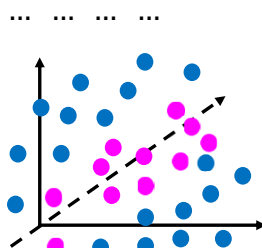
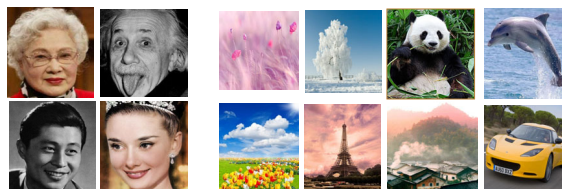
➤ 人脸在特征空间中分布相对集中

4.1 Fisher线性判别动机

训练(Training)



➤ 图像是特征空间中的一个“点”



核心思想

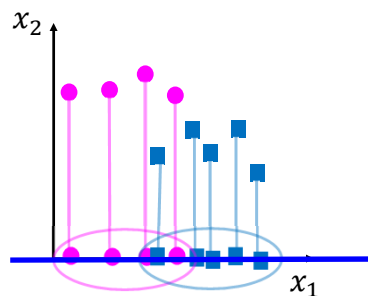
捕获人脸关键特征，将其压缩到低维空间中去。

4.1 Fisher线性判别动机

投影表示:

$$w^T x = \|x\| \|w\| \cos \theta$$

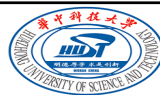
将 x 向 w 投影
(Project x to w)



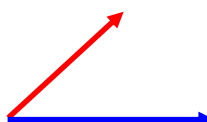
不是好的投影
(Poor Projection)

$$g(x) = w^T x$$

4.1 Fisher线性判别动机

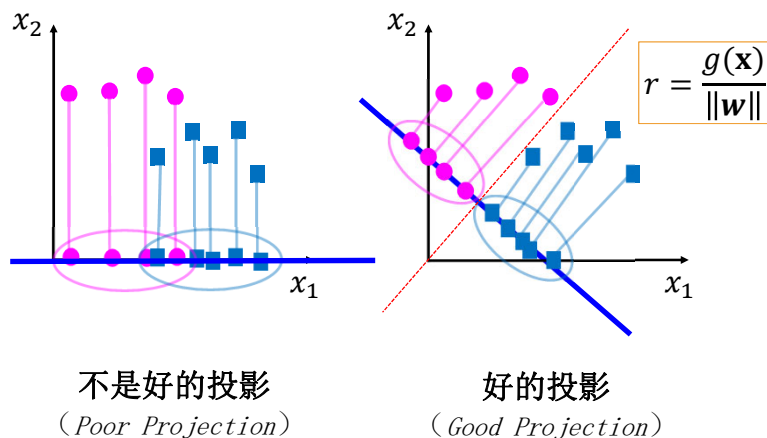


投影表示:

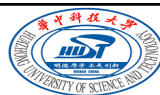


$$w^T x = \|x\| \|w\| \cos \theta$$

将 x 向 w 投影
(Project x to w)



4.1 Fisher线性判别动机

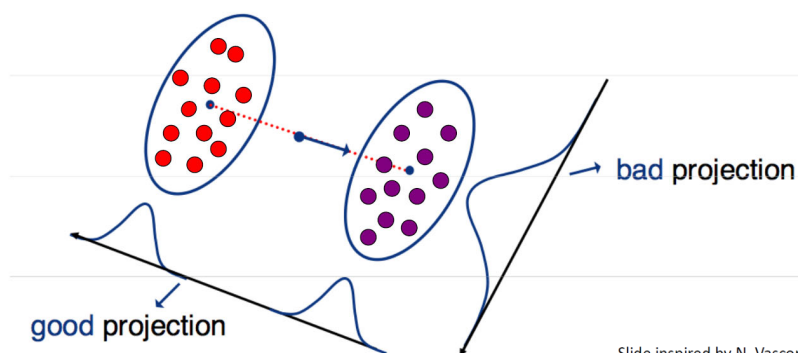


投影表示:



$$w^T x = \|x\| \|w\| \cos \theta$$

将 x 向 w 投影
(Project x to w)



Slide inspired by N. Vasconcelos

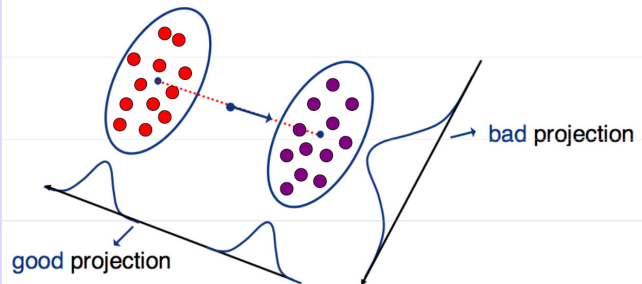
Fisher线性判别的目的: 在两个类别之间**找到最好的区分**
(find the best separation between two classes)

4.1 Fisher线性判别动机



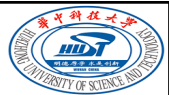
Fisher线性判别的目的:

- 在尽可能保留类别可区分性的前提下**实现降维**
(Perform dimensionality reduction "while preserving as much of the class discriminatory information as possible".)
- 找到让类别最好区分的**投影方向**
(Seeks to find directions along which the classes are best separated.)
- 同时考虑**类内**散布和**类间**散布
(Takes into consideration the scatter within-classes but also the scatter between-classes.)



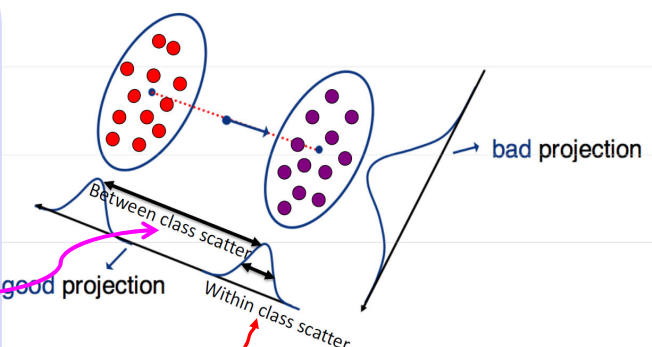
Slide inspired by N. Vasconcelos

4.1 Fisher线性判别动机



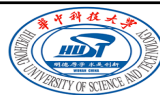
Fisher线性判别的目的:

- 在尽可能保留类别可区分性的前提下实现维数减少
(Perform dimensionality reduction "while preserving as much of the class discriminatory information as possible".)
- 找到让类别最好区分的投影方向
(Seeks to find directions along which the classes are best separated.)
- 同时考虑**类内**散布和**类间**散布
(Takes into consideration the scatter within-classes but also the scatter between-classes.)



Slide inspired by N. Vasconcelos

第四讲 Fisher线性判别 (*Fisher Discriminant Analysis*)

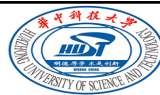


4.1 Fisher线性判别动机 (*The goal of Fisher Linear Discriminant*)

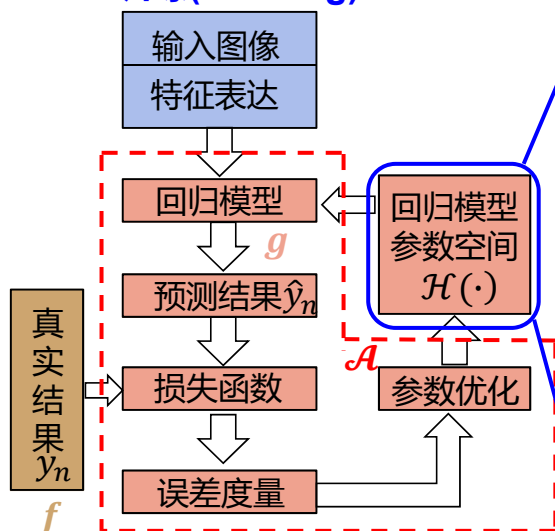
4.2 Fisher线性判别分析 (*Fisher Discriminant Analysis*)

4.3 Fisher线性判别算法 (*Fisher Discriminant Algorithm*)

3.1 线性回归问题



训练(Training)



$\mathcal{H}(\cdot)$ 像什么?

输入: $\mathbf{x} = (x_1, x_2, \dots, x_d)^T$

计算输入 $\mathbf{x}$ 的加权和: $\hat{y} = \sum_{i=1}^d w_i x_i$

回归模型的参数空间: $h(\cdot) \in \mathcal{H}(\cdot)$

$$h(\mathbf{x}) = \mathbf{w}^T \mathbf{x}$$

线性回归的目的: 找到误差最小的拟合模型

4.2 Fisher线性判别分析

二分类问题的Fisher线性判别:

学习最佳投影 $\mathbf{w}^*$ ，它能将所有样本投影到 $\mathbf{w}^*$ 的方向

假设 $s = \mathbf{w}^T \mathbf{x}$ $\mathbf{x} \in \mathcal{R}^d$, $s \in \mathcal{R}^1$

类别集合: $\mathcal{C} = \{c | (1, -1)\}$

第 c 个类别的均值为: $\mu_c = E[\mathbf{x} | y = c] = \frac{1}{N_c} \sum_{n=1}^{N_c} [\mathbf{x}_n | y = c]$

第 c 个类别的协方差为: $\Sigma_c = E[(\mathbf{x} - \mu_c)(\mathbf{x} - \mu_c)^T | y = c]$
 $= \sum_{n=1}^{N_c} [(\mathbf{x}_n - \mu_c)(\mathbf{x}_n - \mu_c)^T | y = c]$

4.2 Fisher线性判别分析

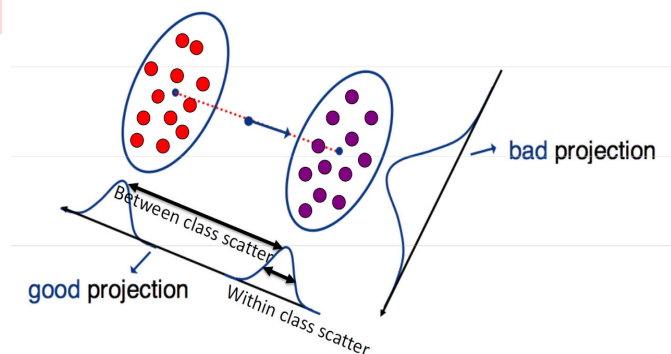
二分类问题的Fisher线性判别:

学习最佳投影 $\mathbf{w}^*$ 的目标函数:

$$J(\mathbf{w}) = \frac{\text{between class scatter}}{\text{within class scatter}}$$

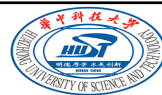
$$\mathbf{w}^* = \underset{\mathbf{w}}{\operatorname{argmax}} J(\mathbf{w})$$

$$J(\mathbf{w}) = \frac{(E[s|y=1] - E[s|y=-1])^2}{\operatorname{var}[s|y=1] + \operatorname{var}[s|y=-1]}$$



Slide inspired by N. Vasconcelos

4.2 Fisher线性判别分析



二分类问题的Fisher线性判别:

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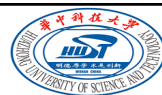
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$$J(\mathbf{w}) = \frac{(E[s|y=1] - E[s|y=-1])^2}{\operatorname{var}[s|y=1] + \operatorname{var}[s|y=-1]}$$

$$\begin{aligned} & (E[s|y=1] - E[s|y=-1])^2 \\ &= (E[\mathbf{w}^T \mathbf{x}|y=1] - E[\mathbf{w}^T \mathbf{x}|y=-1])^2 \\ &= (\mathbf{w}^T (E[\mathbf{x}|y=1] - E[\mathbf{x}|y=-1]))^2 \\ &= (\mathbf{w}^T (\boldsymbol{\mu}_1 - \boldsymbol{\mu}_{-1}))^2 \\ &= \mathbf{w}^T (\boldsymbol{\mu}_1 - \boldsymbol{\mu}_{-1}) (\boldsymbol{\mu}_1 - \boldsymbol{\mu}_{-1})^T \mathbf{w} \end{aligned}$$

4.2 Fisher线性判别分析



二分类问题的Fisher线性判别:

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$$\begin{aligned} \operatorname{var}[s|y=c] &= E[(s - E[s|y=c])^2] \\ &= E[(\mathbf{w}^T \mathbf{x} - E[\mathbf{w}^T \mathbf{x}|y=c])^2] \\ &= E[(\mathbf{w}^T (\mathbf{x} - E[\mathbf{x}|y=c]))^2] \\ &= E[(\mathbf{w}^T (\mathbf{x} - \boldsymbol{\mu}_c))^2] \\ &= E[\mathbf{w}^T (\mathbf{x} - \boldsymbol{\mu}_c) (\mathbf{x} - \boldsymbol{\mu}_c)^T \mathbf{w}] \\ &= \mathbf{w}^T E[(\mathbf{x} - \boldsymbol{\mu}_c) (\mathbf{x} - \boldsymbol{\mu}_c)^T] \mathbf{w} \\ &= \mathbf{w}^T \boldsymbol{\Sigma}_c \mathbf{w} \end{aligned}$$

4.2 Fisher线性判别分析

二分类问题的Fisher线性判别:

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4.2 Fisher线性判别分析

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$$J(\mathbf{w}) = \frac{\mathbf{w}^T (\boldsymbol{\mu}_1 - \boldsymbol{\mu}_{-1}) (\boldsymbol{\mu}_1 - \boldsymbol{\mu}_{-1})^T \mathbf{w}}{\mathbf{w}^T \boldsymbol{\Sigma}_1 \mathbf{w} + \mathbf{w}^T \boldsymbol{\Sigma}_{-1} \mathbf{w}}$$

4.2 Fisher线性判别分析

二分类问题的Fisher线性判别:

学习最佳投影 $\mathbf{w}^*$ 的目标函数:

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$$J(\mathbf{w}) = \frac{\mathbf{w}^T (\boldsymbol{\mu}_1 - \boldsymbol{\mu}_{-1}) (\boldsymbol{\mu}_1 - \boldsymbol{\mu}_{-1})^T \mathbf{w}}{\mathbf{w}^T \boldsymbol{\Sigma}_1 \mathbf{w} + \mathbf{w}^T \boldsymbol{\Sigma}_{-1} \mathbf{w}}$$

$$\mathbf{S}_B = (\boldsymbol{\mu}_1 - \boldsymbol{\mu}_{-1}) (\boldsymbol{\mu}_1 - \boldsymbol{\mu}_{-1})^T$$

$$\mathbf{S}_W = \boldsymbol{\Sigma}_1 + \boldsymbol{\Sigma}_{-1}$$

4.2 Fisher线性判别分析

二分类问题的Fisher线性判别:

学习最佳投影 $\mathbf{w}^*$ 的目标函数:

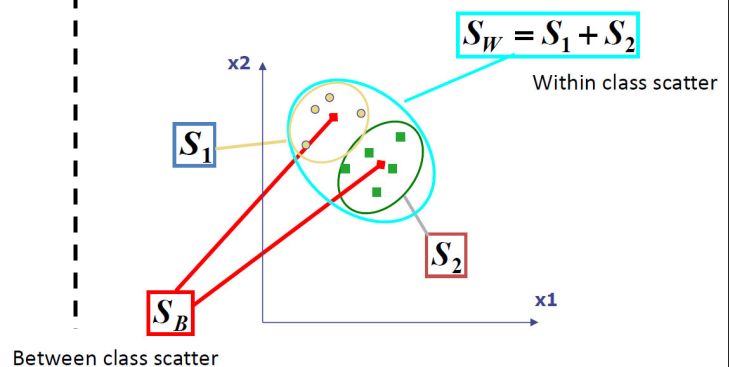
$$J(\mathbf{w}) = \frac{\text{between class scatter}}{\text{within class scatter}}$$

$$\mathbf{w}^* = \underset{\mathbf{w}}{\operatorname{argmax}} J(\mathbf{w})$$

$$J(\mathbf{w}) = \frac{\mathbf{w}^T \mathbf{S}_B \mathbf{w}}{\mathbf{w}^T \mathbf{S}_W \mathbf{w}}$$

$$\mathbf{S}_B = (\boldsymbol{\mu}_1 - \boldsymbol{\mu}_{-1}) (\boldsymbol{\mu}_1 - \boldsymbol{\mu}_{-1})^T$$

$$\mathbf{S}_W = \boldsymbol{\Sigma}_1 + \boldsymbol{\Sigma}_{-1} = \mathbf{S}_1 + \mathbf{S}_2$$



4.2 Fisher线性判别分析

二分类问题的Fisher线性判别:

学习最佳投影 $\mathbf{w}^*$ 的目标函数:

$$J(\mathbf{w}) = \frac{\text{between class scatter}}{\text{within class scatter}}$$

$$\mathbf{w}^* = \underset{\mathbf{w}}{\operatorname{argmax}} J(\mathbf{w})$$

$$J(\mathbf{w}) = \frac{\mathbf{w}^T \mathbf{S}_B \mathbf{w}}{\mathbf{w}^T \mathbf{S}_w \mathbf{w}}$$

最大化目标函数问题转化为:

$$\max_{\mathbf{w}} \mathbf{w}^T \mathbf{S}_B \mathbf{w} \quad \text{Subject to } \mathbf{w}^T \mathbf{S}_w \mathbf{w} = K$$

利用拉格朗日乘子法 (Lagrange multipliers) :

$$L(\mathbf{w}, \lambda) = \mathbf{w}^T \mathbf{S}_B \mathbf{w} + \lambda(K - \mathbf{w}^T \mathbf{S}_w \mathbf{w})$$

$$= \mathbf{w}^T (\mathbf{S}_B - \lambda \mathbf{S}_w) \mathbf{w} + \lambda K$$

$$\nabla L_w(\mathbf{w}, \lambda) = \frac{\partial L(\mathbf{w}, \lambda)}{\partial \mathbf{w}} = \mathbf{0}^T$$

$$2(\mathbf{S}_B - \lambda \mathbf{S}_w) \mathbf{w} = \mathbf{0} \Rightarrow \mathbf{S}_B \mathbf{w} = \lambda \mathbf{S}_w \mathbf{w}$$

4.2 Fisher线性判别分析

二分类问题的Fisher线性判别:

学习最佳投影 $\mathbf{w}^*$ 的目标函数:

$$J(\mathbf{w}) = \frac{\text{between class scatter}}{\text{within class scatter}}$$

$$\mathbf{w}^* = \underset{\mathbf{w}}{\operatorname{argmax}} J(\mathbf{w})$$

$$J(\mathbf{w}) = \frac{\mathbf{w}^T \mathbf{S}_B \mathbf{w}}{\mathbf{w}^T \mathbf{S}_w \mathbf{w}}$$

$$\max_{\mathbf{w}} \mathbf{w}^T \mathbf{S}_B \mathbf{w} \quad \text{Subject to } \mathbf{w}^T \mathbf{S}_w \mathbf{w} = K$$

$$L(\mathbf{w}, \lambda) = \mathbf{w}^T \mathbf{S}_B \mathbf{w} + \lambda(K - \mathbf{w}^T \mathbf{S}_w \mathbf{w})$$

$$\nabla L_w(\mathbf{w}, \lambda) = \mathbf{0}$$

$$\mathbf{S}_B \mathbf{w} = \lambda \mathbf{S}_w \mathbf{w}$$

如果 $\mathbf{S}_w^{-1} = (\boldsymbol{\Sigma}_1 + \boldsymbol{\Sigma}_{-1})^{-1}$ 存在, 则有:

$$\mathbf{S}_w^{-1} \mathbf{S}_B \mathbf{w} = \lambda \mathbf{w}$$

$$\mathbf{S}_w^{-1} (\boldsymbol{\mu}_1 - \boldsymbol{\mu}_{-1}) (\boldsymbol{\mu}_1 - \boldsymbol{\mu}_{-1})^T \mathbf{w} = \lambda \mathbf{w}$$

$$\mathbf{S}_w^{-1} (\boldsymbol{\mu}_1 - \boldsymbol{\mu}_{-1}) a = \lambda \mathbf{w} \quad (\boldsymbol{\mu}_1 - \boldsymbol{\mu}_{-1})^T \mathbf{w} = a \text{ 标量}$$

$$\mathbf{S}_w^{-1} (\boldsymbol{\mu}_1 - \boldsymbol{\mu}_{-1}) = \frac{\lambda}{a} \mathbf{w}$$

只关注投影向量的方向 :

$$\mathbf{w}^* = \mathbf{S}_w^{-1} (\boldsymbol{\mu}_1 - \boldsymbol{\mu}_{-1})$$

4.2 Fisher线性判别分析

二分类问题的Fisher线性判别:

学习最佳投影 $\mathbf{w}^*$ 的目标函数:

$$J(\mathbf{w}) = \frac{\text{between class scatter}}{\text{within class scatter}}$$

$$\mathbf{w}^* = \underset{\mathbf{w}}{\operatorname{argmax}} J(\mathbf{w})$$

$$J(\mathbf{w}) = \frac{\mathbf{w}^T \mathbf{S}_B \mathbf{w}}{\mathbf{w}^T \mathbf{S}_W \mathbf{w}}$$

$$\max_{\mathbf{w}} \mathbf{w}^T \mathbf{S}_B \mathbf{w} \quad \text{Subject to } \mathbf{w}^T \mathbf{S}_W \mathbf{w} = K$$

$$L(\mathbf{w}, \lambda) = \mathbf{w}^T \mathbf{S}_B \mathbf{w} - \lambda(\mathbf{w}^T \mathbf{S}_W \mathbf{w} - K)$$

$$\nabla L_{\mathbf{w}}(\mathbf{w}, \lambda) = \mathbf{0} \Rightarrow \mathbf{w}^* = \mathbf{S}_W^{-1}(\mu_1 - \mu_{-1})$$

找到投影向量后, 对任一测试样本 $\mathbf{x}$:

$$s = \mathbf{w}^{*T} \mathbf{x} = (\mathbf{S}_W^{-1}(\mu_1 - \mu_{-1}))^T \mathbf{x}$$

假设类别的判别门限设为 s' :

$$s' = \frac{\mathbf{w}^{*T}(\mu_1 + \mu_{-1})}{2}$$

对任一测试样本 $\mathbf{x}$ 所属类别的判断:

$$\begin{cases} y = 1 & \text{if } s = \mathbf{w}^{*T} \mathbf{x} > s' \\ y = -1 & \text{if } s = \mathbf{w}^{*T} \mathbf{x} < s' \end{cases}$$

4.2 Fisher线性判别分析

判别门限:

- 维数d和样本数N极大, 用贝叶斯决策规则
- 不满足上述条件, 使用先验知识确定阈值点

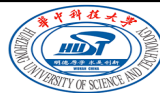
$$s' = \frac{\mathbf{w}^{*T}(\mu_1 + \mu_{-1})}{2}$$

$$y_0^{(1)} = \frac{\tilde{m}_1 + \tilde{m}_2}{2}$$

$$y_0^{(2)} = \frac{N_1 \tilde{m}_1 + N_2 \tilde{m}_2}{N_1 + N_2} = \tilde{m}$$

$$y_0^{(3)} = \frac{\tilde{m}_1 + \tilde{m}_2}{2} + \frac{\ln(P(\omega_1)/P(\omega_2))}{N_1 + N_2 - 2}$$

第四讲 Fisher线性判别 (*Fisher Discriminant Analysis*)

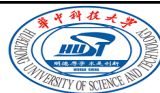


4.1 Fisher线性判别动机 (*The goal of Fisher Linear Discriminant*)

4.2 Fisher线性判别分析 (*Fisher Discriminant Analysis*)

4.3 Fisher线性判别算法 (*Fisher Discriminant Algorithm*)

4.3 Fisher线性判别算法



二分类问题的Fisher线性判别算法:

① 获取具有标签的两类样本

② 依据下式得到 μ_1 和 μ_{-1} :

$$\mu_c = \frac{1}{N_c} \sum_{n=1}^{N_c} [\mathbf{x}_n | y = c]$$

③ 依据下式得到 Σ_1 和 Σ_{-1} :

$$\Sigma_c = \sum_{n=1}^{N_c} [(\mathbf{x}_n - \mu_c)(\mathbf{x}_n - \mu_c)^T | y = c]$$

④ 计算类内总离差阵: $S_w = \Sigma_1 + \Sigma_{-1}$

⑤ 计算类内总离差阵的逆: S_w^{-1}

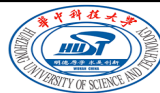
⑥ 计算最佳投影: $\mathbf{w}^* = S_w^{-1}(\mu_1 - \mu_{-1})$

⑦ 计算判别门限 s' : $s' = \frac{\mathbf{w}^{*T}(\mu_1 + \mu_{-1})}{2}$

⑧ 对任一测试样本 $\mathbf{x}$:

$$\begin{cases} y = 1 & \text{if } s = \mathbf{w}^{*T} \mathbf{x} > s' \\ y = -1 & \text{if } s = \mathbf{w}^{*T} \mathbf{x} < s' \end{cases}$$

第四讲 Fisher线性判别 (*Fisher Discriminant Analysis*)



4.1 Fisher线性判别动机

在尽可能保留类别可区分性的前提下实现维数减少

4.2 Fisher线性判别分析

找到让类别最好区分的投影方向

4.3 Fisher线性判别算法

通过计算类内散布和类间散布，找到最佳 w^* 和判别门限 s'